

DEPARTMENT: EMERGING TECHNOLOGIES

AI-Native Immersive Computing: Toward Scalable, Intelligent, and Trustworthy Digital–Physical Continuums

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Immersive computing is emerging as the next-generation computing paradigm, yet current systems are hindered by fragmented multimodal intelligence, siloed design, rigid infrastructure, and insufficient guarantees of trust and application grounding. This article presents a vision for artificial intelligence (AI)-native immersive computing, in which AI is embedded as a first-class primitive throughout the entire system stack. We propose a research agenda structured around three synergistic pillars to bridge these gaps. It prioritizes the creation of multimodal foundation models capable of unifying diverse data representations, performing adaptive inference in real time, and generating realistic 3-D world content. This intelligence is sustained by networked systems and infrastructure that leverage learning-driven orchestration and real-time coordination across scalable, shared platforms. To translate these advancements into meaningful real-world impact, the framework incorporates trustworthy, domain-specific application design by embedding end-to-end privacy, ensuring robust, reliable performance, and centering on task-centric operational requirements.

Computing is undergoing a fundamental transformation, shifting from personal computers and smartphones to distributed cloud/edge platforms. Each paradigm shift has redefined how humans interact with digital content. Today, immersive computing, encompassing augmented reality (AR), virtual reality (VR), and mixed reality (MR), collectively referred to as extended reality (XR), is poised to become the next major paradigm, enabling users to engage seamlessly with digital information embedded within their physical environments.¹ By merging perception, cognition, computation, and communication into

spatial contexts, it promises to fundamentally reshape how we learn, work, and collaborate.

Despite rapid hardware advances, current systems remain far from realizing the full potential of immersive computing. As shown in [Figure 1](#), we identify three critical gaps that motivate a move toward a more co-evolutionary approach. First, existing systems suffer from *fragmented multimodal intelligence*, in which heterogeneous sensory inputs, spanning visual, auditory, and haptic modalities, are processed as isolated data streams rather than synthesized into a unified representation that enables foundation models to reason holistically about the environment. Second, *siloed design and rigid infrastructure* force system optimizations to be reactive, passively responding to workloads rather than actively predicting them. This is particularly problematic for geo-distributed experiences such as telepresence² and teleoperation,³ which require tight coordination

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across the device–edge–cloud spectrum. Third, there is *insufficient trust guarantee and application grounding*. As systems capture rich visual and behavioral data, they create unprecedented risks of data leakage and misuse.⁴ Current security solutions are often isolated “add-ons,” while many applications fail to capture the specific workflows of high-impact domains such as health care,⁵ education,⁶ and scientific discovery.⁷

In parallel, advances in multimodal foundation models⁸ have demonstrated remarkable capabilities in perception and reasoning. However, these have yet to be fully integrated into the core abstractions of immersive computing. Instead, artificial intelligence (AI) is often treated as a peripheral component for specific tasks such as object recognition, rather than a unifying principle for the entire system.

In this article, we argue that overcoming these limitations requires a fundamental shift toward *AI-native immersive computing*, in which AI is embedded as a first-class primitive across the entire system stack. In this paradigm, immersive systems are no longer reactive pipelines but intelligent, adaptive platforms that can understand, predict, and respond to dynamic environments. To realize this, we propose a research agenda structured around three synergistic pillars.

We first focus on creating *multimodal foundation models* capable of unifying diverse data representations, performing adaptive inference in real time, and generating or predicting the evolution of world content. This intelligence is sustained by *networked systems and infrastructure* that leverage learning-driven orchestration and real-time coordination across scalable, shared platforms. Finally, we move to *trustworthy and application-grounded computing* by embedding end-to-end privacy, ensuring robust and reliable performance, and centering on domain-specific design needs. While privacy is often treated as a general system property, we elevate it to be part of a standalone pillar to address the unique tension between the intrusive nature of immersive sensing and the transparency required for AI-native reasoning.

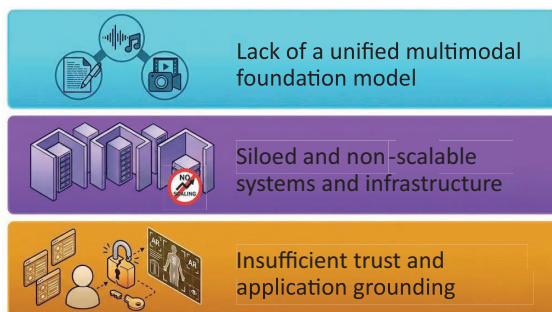


FIGURE 1. Key limitations of current immersive systems.

Central to this vision is a *holistic co-evolution principle*. Advances in foundation models must inform systems and networking, while system-level constraints and application requirements must, in turn, influence model architectures and training objectives. By integrating these layers into a single ecosystem, immersive computing can evolve into a predictive, scalable, and trustworthy digital–physical continuum.

In the remainder of this article, we articulate this vision in detail. We first examine the limitations of current immersive systems and motivate the need for AI-native design. We then present a research agenda structured around three synergistic pillars: multimodal foundation models for the physical world, scalable networked systems and infrastructure, and trustworthy transformative applications. Finally, we discuss open challenges and outline a path toward realizing the envisioned digital–physical continuums.

WHY AI-NATIVE MATTERS?

Despite recent progress in immersive technologies, today's systems remain fundamentally constrained by design principles inherited from earlier computing paradigms. Most of them still follow pipeline-based designs, in which capturing, encoding, transmission, rendering, and interaction are treated as largely independent modules. While this decomposition enables incremental optimization of individual components, it limits the system's ability to reason holistically about complex environments and to adapt in real time to changing conditions. In emerging scenarios of immersive interaction between humans and AI, as shown in [Figure 2](#), current pipelines fail to tightly integrate perception, reasoning, and generation into a unified loop.

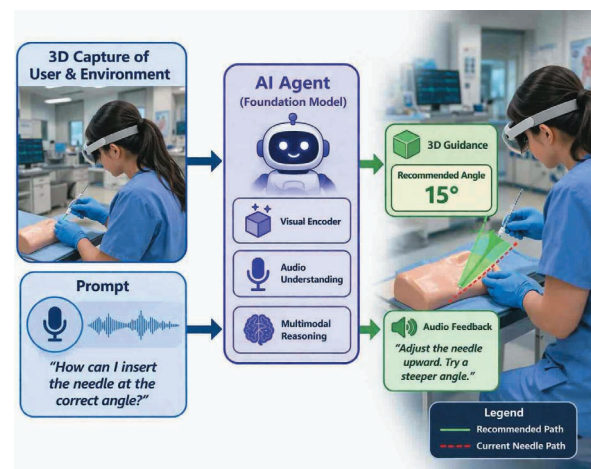


FIGURE 2. Motivational use case: Immersive interaction between human and generative AI for medical training.

These limitations highlight three fundamental gaps that motivate a more integrated, AI-native approach.

Lack of a Unified Multimodal Foundation Model

A primary challenge is the lack of unified representations of the physical world that bridge the gap between raw sensory data and high-level reasoning. In the use case depicted in [Figure 2](#), the system must jointly reason over 3-D human pose, environmental context, and spoken intent to generate context-aware guidance. However, existing approaches process these modalities in isolation, making it difficult to align spatial understanding with high-level intent. This fragmentation prevents the system from forming a coherent “world model”⁹ that understands how physical objects and human actions relate semantically. Without a shared model that captures both physical structure and semantic meaning, systems cannot effectively support predictive behaviors or adapt interaction modalities in ways that feel natural to the user. This fragmented multimodal intelligence also complicates cross-layer optimization, as system and network decisions are made without a deep understanding of the underlying content and context, leading to inefficient resource utilization and a lack of situational awareness.

Siloed, Nonscalable Systems and Infrastructure

Equally critical is the limitation of reactive and loosely coordinated system management, which fails to meet the intensive demands of next-generation XR. Traditional approaches rely on predefined heuristics or short-term feedback loops to handle variability in compute and network conditions. However, immersive applications demand stringent latency and synchronization requirements,² particularly when real-time AI reasoning and 3-D content generation are required. In health-care training, for instance, even minor delays in generating 3-D overlays can break immersion or directly affect safety-critical performance. Yet current systems treat model inference and content delivery as separate, siloed steps without predictive coordination, limiting scalability in multiuser scenarios. Furthermore, the absence of shared, scalable infrastructure hinders the community’s ability to evaluate end-to-end systems at scale, as researchers are often forced to build custom, noninteroperable testbeds.¹⁰ What is needed are learning-driven mechanisms and platforms that can proactively allocate resources across the device–edge–cloud spectrum, ensuring robust performance

even as the number of users and the complexity of AI tasks grow.

Insufficient Trust and Application Grounding

Immersive computing introduces unprecedented risks regarding privacy, security, and trust that current frameworks are ill-equipped to handle. Systems continuously capture rich 3-D representations of physical spaces and behavioral cues from users, creating significant potential for data leakage and the unauthorized reconstruction of sensitive environments.⁴ Current solutions typically address these concerns in isolation, such as simple encryption at the network layer, and lack a holistic, system-wide approach to trustworthy intelligence that protects data from capture to inference. Simultaneously, most existing systems fail to account for *domain-specific requirements* and user workflows in high-stakes areas such as health care, education, and scientific discovery. Without incorporating such human-centric considerations, immersive systems risk providing technically advanced yet practically limited solutions that do not fit into professional environments. Addressing this gap requires embedding end-to-end trust mechanisms directly into the system architecture and grounding system design in practical constraints, safety protocols, and expert workflows of real-world use cases.

In summary, these persistent limitations necessitate a paradigm shift toward AI-native design. Only by integrating intelligence into the foundational architecture can immersive systems move beyond reactive pipelines and evolve into the adaptive, interactive ecosystems required for next-generation computing.

RESEARCH AGENDA

Realizing the vision of AI-native immersive computing requires a coordinated research effort that unifies advances in AI, systems, networking, infrastructure, privacy, and human-centered application design. We outline a research agenda organized around three synergistic pillars, as shown in [Figure 3](#), each addressing a critical limitation of current approaches and reinforcing the others through a holistic co-evolution principle presented in the “Holistic Co-Evolution Principle” section.

Multimodal Foundation Models for Physical World

At the core of AI-native immersive computing lies the need for *unified, multimodal representations* that can seamlessly bridge the gap between raw sensory

perception and deep semantic reasoning. Current immersive systems typically rely on fragmented data structures, disjoint 3-D meshes for spatial geometry, discrete labels for semantics, and separate skeletal models for human motion, which leads to fundamental inconsistencies in dynamic environments. For instance, a system might struggle to align a user's verbal intent with the precise physical coordinates of a 3-D object in real time, causing interaction lag and reduced immersion. Foundational research must therefore shift toward models that encode geometry, semantics, and temporal dynamics into a coherent, shared latent space. By establishing this high-fidelity "digital twin," the system can reason not only about what exists in a scene but also about the underlying physics of object motion and the causal dependencies between users and their surroundings. Such representations serve as the bedrock of a unified world model in which heterogeneous sensory streams are synthesized into a single, actionable intelligence layer.

This pillar also demands a fundamental shift in execution toward *real-time, adaptive inference* and online learning. Unlike traditional large language or vision models trained on massive, static offline datasets, AI-native immersive models must continuously update their parameters and inference strategies based on high-bandwidth, streaming multimodal data. This raises significant challenges in online learning efficiency and model compression, as the architecture must balance high-fidelity reasoning with the stringent resource constraints and thermal envelopes of wearable devices. Achieving this goal requires new neural architectures that support incremental updates without catastrophic forgetting, allowing the system to adapt

to new environments or specific user behaviors on the fly. By dynamically partitioning model weights across the device–edge–cloud continuum, the system can ensure that the most latency-sensitive reasoning occurs locally, while complex, global updates are synchronized via the back end, effectively enabling the digital–physical continuum to respond instantaneously to real-world stimuli.

Beyond perception and adaptation, these models must provide robust *generative and predictive capabilities* to mask the inherent latencies of distributed systems. By anticipating future states of the environment or predicting user trajectories, the system can proactively generate and render immersive content, ensuring a seamless experience even under fluctuating network or computing conditions.¹¹ This enables a new class of generative interaction modalities where the system synthesizes missing sensor data, performs semantic compression of 3-D scenes for efficient transmission, and creates personalized guidance overlays. For example, in a complex assembly task, the model could predict a user's intent and generate a 3-D visualization of the correct component placement before the user even reaches for the part. Ultimately, this research direction aims to establish a framework where the boundaries between sensing, reasoning, and content generation disappear, providing a unified foundation that transforms immersive environments from reactive displays into proactive, intelligent partners.

Scalable Networked Systems and Infrastructure

To sustain complex foundation models at scale, the underlying systems and infrastructure must be

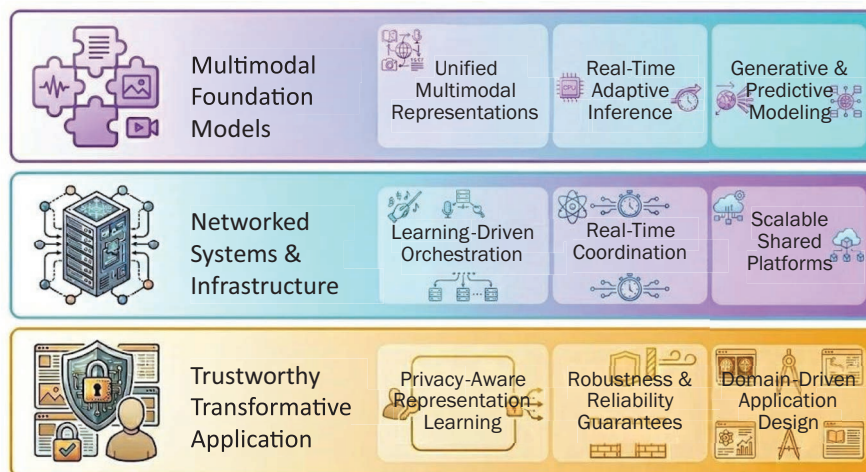


FIGURE 3. Proposed research agenda for AI-native immersive computing.

fundamentally rethought as an intelligent, unified *device-edge-cloud fabric*. This pillar merges the technical challenges of distributed systems design with the necessity for community-driven experimentation platforms. Traditional systems treat AI as a modular add-on at the application layer, but true AI-native immersive computing requires tight integration between learning mechanisms and core system abstractions. A central challenge is developing *learning-driven orchestration* and resource management. Instead of relying on static heuristics or reactive feedback loops, the system should leverage the predictive insights from foundation models to proactively allocate compute, memory, and bandwidth. By anticipating workload spikes and network variability before they occur, the fabric can meet the submillisecond synchronization requirements necessary for multiuser immersive experiences.¹² This involves adaptive partitioning of AI tasks, deciding in real time which components of a world model should execute on the user's headset for immediate feedback, which should be offloaded to a nearby edge server for spatial mapping, and which require the massive scale of a centralized cloud for global state coordination.

Supporting these experiences requires *real-time coordination* that maintains strict temporal and spatial synchronization across heterogeneous environments. For co-located users, the system must ensure that shared digital artifacts are anchored to the same physical coordinates with subcentimeter precision, necessitating low-latency, peer-to-peer spatial alignment and consistent haptic-visual feedback loops. In geo-distributed collaborative settings, the challenge shifts to managing high-frequency updates of environmental/content changes while masking the effects of geographic jitter and packet loss. To maintain a seamless shared state, the system must employ intelligent delta-tracking and speculative execution of environment dynamics, ensuring that remote interactions feel instantaneous and coherent across all users.¹³ This research direction aims to establish adaptive protocols that are aware of 3-D content and user intent, ensuring that the shared state remains coherent and interactive whether users are in the same room or across the globe.

Enabling real-time, multiuser immersive experiences across geographically distributed locations introduces additional hurdles in scalability. AI-native approaches can mitigate these issues by using semantic compression to transmit only the most critical updates and by using local generative models to “fill in” the gaps caused by packet loss.² Beyond runtime

management, this pillar addresses the critical need for *shared, community-driven infrastructure* to accelerate reproducible innovation. The current lack of standardized platforms often forces researchers to build custom, siloed testbeds that are difficult to compare or scale.¹⁰ We propose developing extensible toolkits and standardized benchmarks¹⁴ for AI-native workloads that incorporate realistic multiuser scenarios and diverse network conditions. By providing the research community with representative datasets and open source deployment frameworks, this pillar establishes a common foundation for large-scale experimentation.

Trustworthy Transformative Applications

The final pillar grounds these technical advances in the practical constraints of real-world use cases while treating *end-to-end trust and application grounding* as fundamental system requirements. The decision to elevate privacy and trust as part of a standalone pillar is necessitated by the inherent intimacy of immersive data. We emphasize that trust must be a first-class design objective that shapes the architecture of models and networks, rather than a constraint applied after the system is built.

Unlike traditional systems that process discrete data packets, immersive frameworks capture high-resolution digital shadows of the physical world, including 3-D spatial layouts of private spaces and biometric markers.¹⁵ This creates a fundamental conflict: AI-native capabilities require access to raw sensory streams, while user safety requires strict confidentiality. Addressing this challenge requires moving beyond reactive security patches toward a comprehensive approach that treats privacy as a cross-layer property. Research must focus on selective sensing and *privacy-aware representation learning*, ensuring that only the information strictly necessary for the application is processed and that sensitive data are anonymized through on-device processing before it ever reaches the network. These protection mechanisms must be designed to be computationally efficient, providing strong security guarantees, such as differential privacy or secure enclave execution, without compromising the real-time responsiveness that immersive interaction demands. Beyond training, AI-native design should ensure privacy during inference via local semantic extractors. By transforming raw pixels into abstract, anonymized representations on-device, the system processes only intent rather than sensitive visual data, making privacy an inherent architectural byproduct.

Equally vital is the advancement of *secure computation and robust AI model design* to protect against both unintentional failures and malicious exploitation. Techniques such as privacy-preserving federated learning can help safeguard data during processing,¹⁵ but they must be optimized to meet the submillisecond latency and high-throughput performance requirements of immersive workloads. Furthermore, as these systems move from controlled labs to diverse, potentially untrusted environments, models must be inherently robust to adversarial inputs and system-level attacks that could manipulate 3-D perception or bypass privacy filters. Developing these reliability guarantees requires designing self-healing architectures for AI-driven rendering and interaction loops, ensuring that even under network instability or edge-case sensory inputs, the system maintains a predictable and safe state through real-time failure mitigation. This research direction seeks to build a resilient foundation in which user trust is earned through verifiable stability and an ironclad defense against the evolving digital threat landscape.

Simultaneously, the impact of these systems depends on their ability to move beyond generic functionalities and provide *domain-specific application grounding*. Current XR systems prioritize rendering quality or basic latency, but they often fail to account for the unique workflows and contextual constraints of high-stakes fields such as health care, education, and scientific discovery.¹⁶ For example, in a medical

training scenario, effective assistance requires not just accurate 3-D visualization, but a deep understanding of surgical protocols, patient safety constraints, and the user's specific progression through a task.⁵ In these high-stakes domains, we distinguish between privacy (confidentiality) and reliability (systemic trust). AI-native reliability enhances fault tolerance by using predictive models to mask hardware or network failures. For example, during remote surgical training, if a packet is dropped, local generative models provide "semantic persistence" by predictively rendering instrument motion ("Scalable Networked Systems and Infrastructure" section). This ensures continuous visual feedback and system trustworthiness, leveraging AI to bridge performance gaps that classical systems cannot.

HOLISTIC CO-EVOLUTION PRINCIPLE

A defining principle of the envisioned AI-native immersive computing is *holistic co-evolution*, a paradigm shift where multimodal foundation models, networked systems and infrastructure, and trustworthy transformative applications are no longer treated as independent modules but as tightly integrated entities linked through continuous, bidirectional feedback loops, as shown in Figure 4. Together, they form a fully connected and dynamic design space where each pillar both informs and is radically reshaped by the requirements

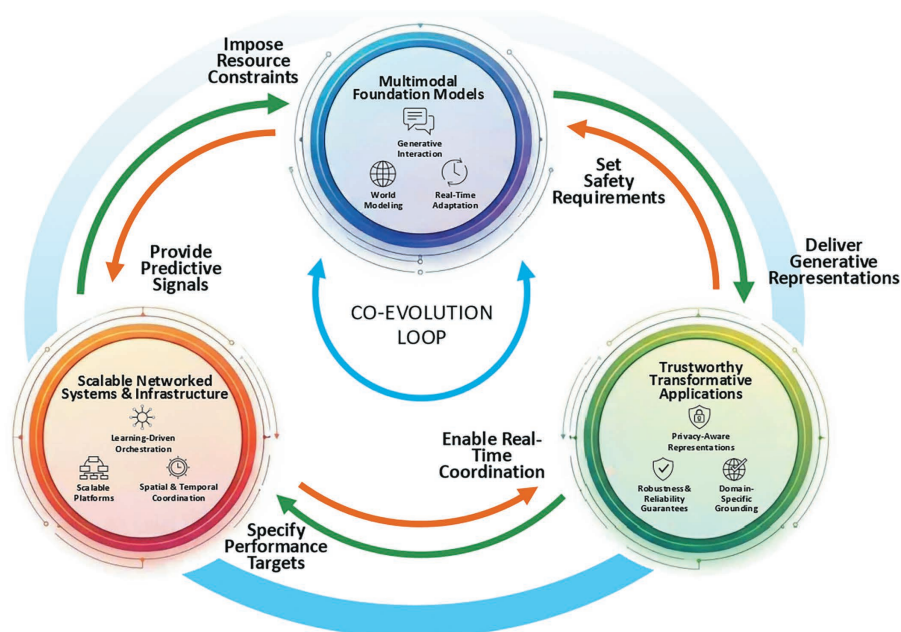


FIGURE 4. Holistic co-evolution: A defining principle of AI-native immersive computing.

and capabilities of the others. This approach moves away from the rigid, one-way optimizations of traditional computing pipelines toward a fluid ecosystem where intelligence is embedded at every layer of the stack.

The first trajectory of this co-evolution centers on how *multimodal foundation models* serve as the primary intelligence engine for the entire stack, providing the high-level reasoning that replaces traditional, hard-coded heuristics. These models provide rich, semantically aware predictive signals that enable *networked systems and infrastructure* to transition from reactive to proactive management, enabling anticipatory bandwidth allocation and intelligent task offloading based on predicted user trajectories and environmental dynamics. By understanding the intent behind a user's movement, the model can signal the network to prefetch 3-D assets or warm up edge compute nodes before the data are explicitly requested. Simultaneously, these models empower *trustworthy transformative applications* by offering unified world representations and generative interaction capabilities that can "fill in" missing sensory data or provide context-aware guidance overlays. By synthesizing disparate sensory streams, including audio, vision, and haptics, into a coherent latent space, the foundation models provide the deep contextual reasoning necessary to ensure that application-level interactions are both semantically meaningful and spatially accurate, even when operating under degraded sensor conditions or in highly cluttered physical environments.

Conversely, the constraints and capabilities of *networked systems and infrastructure* fundamentally dictate the feasibility, architecture, and deployment strategies of the other two pillars. The availability of compute resources, storage hierarchies, and network variability across the device-edge-cloud fabric defines the practical "design envelope" for *multimodal foundation models*, thereby forcing model architectures to be inherently adaptive and computationally efficient for real-time inference on heterogeneous, resource-constrained hardware. This infrastructure layer determines where a model is partitioned and how its weights are updated, directly impacting the model's accuracy and responsiveness. For *trustworthy transformative applications*, the infrastructure provides the essential backbone for secure execution environments, trusted enclaves, and low-latency coordination protocols that ensure robust, reliable multi-user synchronization. By exposing low-level system metrics, such as battery life, link quality, and data locality, the infrastructure layer compels models to be

context-aware and applications to be resource elastic, ensuring that the user experience remains stable and fluid despite the inherent volatility and unpredictability of distributed physical environments.

Finally, *trustworthy transformative applications* ground the entire ecosystem by supplying the user-centric requirements and domain-specific workloads that steer the technical evolution of the underlying layers. These applications define safety-critical constraints, such as subcentimeter overlay precision in medical training or ironclad data confidentiality in sensitive educational settings, that force *multimodal foundation models* to prioritize specific semantic features and adopt privacy-aware representation-learning techniques from the outset. Rather than being an afterthought, these application-layer demands dictate the precise performance targets for *networked systems and infrastructure*, driving the development of specialized orchestration policies that prioritize task-specific synchronization and reliability over generic throughput. This bidirectional interaction ensures that the technical advancements in models and systems remain strictly aligned with the practical, human-centered needs of high-stakes domains, preventing the technology from diverging from its intended transformative impact and ensuring that the convergence of digital and physical worlds remains productive, safe, and intuitively accessible to the end user.

OPEN PROBLEMS AND RESEARCH OPPORTUNITIES

While AI-native immersive computing offers a compelling vision, realizing it in practice poses fundamental open problems spanning coordinated scalability, generative reliability, and recursive system stability, as shown in Figure 5. Addressing these challenges is essential to ensure that immersive systems are not only powerful but also responsibly autonomous.

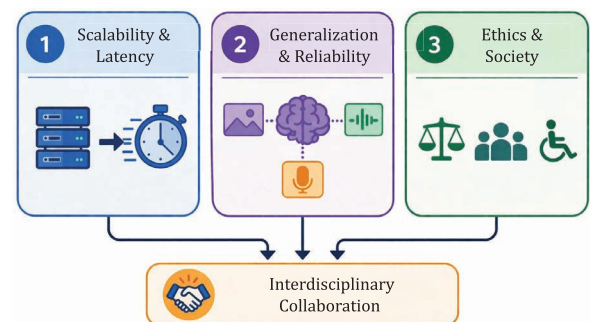


FIGURE 5. Open problems and research opportunities toward AI-native immersive computing.

A primary challenge lies in the semantic-latency paradox. Immersive applications require submillisecond response times,² yet the AI-native approach of using deep semantic extractors to minimize data transmission introduces a “compute-latency” penalty. While classical systems struggle with raw throughput, AI-native systems must balance the overhead of on-device model inference against the time saved by transmitting abstract representations. Resolving this requires a new class of asymmetric architectures where semantic extraction is lightweight to be truly instantaneous while remaining rich for back-end global state coordination.

Another critical challenge concerns generative reliability and compositional safety. Unlike traditional deterministic systems, AI-native systems use foundation models to predictively fill gaps in synchronization. This introduces the risk of “semantic hallucinations,”¹⁷ where a model renders a physically plausible but factually incorrect state (e.g., a misplaced virtual surgical guide during a network jitter event). Ensuring safety in these generative loops is fundamentally different from classical fault tolerance; it requires new formal verification methods¹⁸ that can certify the accuracy of AI-driven rendering in safety-critical, real-time environments.

Beyond technical performance, the co-evolution of these pillars raises unique ethical¹⁹ and recursive stability issues. In a tightly integrated fabric, a model’s biased prediction could lead to faulty infrastructure orchestration, which in turn reinforces biased data collection. This recursive feedback loop is a challenge unique to the holistic co-evolution principle and could lead to systemic instability or model collapse if not properly regulated. Addressing this requires not only bias mitigation but also the development of stability guarantees for interconnected, learning driven control loops.

These challenges highlight the need for a holistic research paradigm that transcends traditional disciplinary boundaries. Addressing the complex, interconnected issues inherent in immersive computing requires the tight integration of foundational intelligence with secure, networked infrastructure and human-centered design principles.²⁰ By bridging the gap between technical innovation and societal impact, researchers have a rich opportunity to shape immersive computing into a truly transformative and responsible ecosystem.

CONCLUSION

AI-native immersive computing represents a fundamental shift from fragmented, reactive pipelines to intelligent, scalable, adaptive, and trustworthy digital-physical continuums that seamlessly integrate models, systems, networking, trust, infrastructure, and

applications. By treating AI as a first-class primitive to enable holistic co-evolution, this paradigm can unlock real-time, privacy-preserving, and collaborative immersive experiences. While significant challenges remain in areas including scalability, generalization, and ethics, the opportunities for transformation in domains such as education, health care, and scientific discovery are profound. We believe that advancing this integrated approach will establish the foundation for the next-generation human-centered computing systems.

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